Fernando Martinez-Lopez

f.martinezlpz@gmail.com 929-374-8678 New York, NY FernandoML.me fernando-ml
https://www.linkedin.com/in/mlfernando Google Scholar: Fernando Martinez (Fordham)

EDUCATION

Ph.D. in Computer Science Aug 2024 – Expected May 2029 | New York, USA
Fordham University
Research: Reinforcement Learning, World Models, Representation Learning, Multi-Agent Learning, and Game/Decision Theory
Roles: Research & Teaching Assistant

Master in Data Science - MSc Jan 2022 – May 2023 | New York, USA
Fordham University
GPA: 4.00 / 4.00
Relevant Courses: Cloud Computing, Big Data Computing, Blockchain Technology, Data Mining, Machine Learning, and Deep Learning.
Roles: Research & Teaching Assistant (CISC 5835 Algorithms for Data Science & CISC 6745 Data Visualization)

Master of Business Administration - MBA Mar 2021 – Mar 2022 | Washington, D.C., USA
Quantic School of Business and Technology
Grade: Honors

B.S. in Business Engineering Aug 2014 – Apr 2018 | Santo Domingo, DR
Santo Domingo Institute of Technology (INTEC)
GPA: 3.81 / 4.00 | Summa Cum Laude
INTEC Meritorious Scholarship (top 1% program)

INDUSTRY EXPERIENCE

Fordham-IBM Research Intern May 2025 – Jul 2025 | New York, USA
Fordham University
- Conducted collaborative research on coupling online representation learning and reinforcement learning through a Stackelberg-game formulation and bilevel optimization, resulting in co-authoring "Stackelberg Coupling of Online Representation Learning and Reinforcement Learning," published at ICLR 2026; first author.

Quantitative Modeling - AVP MUSA CCAR Capital Planning Jul 2023 – Aug 2024 | New York, USA
MUFG Americas
- Led the development and implementation of the CCAR Capital Planning interconnected model suite framework, enhancing operational efficiency through automation and significantly reducing the risk of human errors.
- Redeveloped CCAR quantitative models to improve performance and deploy new desk strategies, while upholding robustness and business soundness.
- Conduct model validation, performance monitoring, and stress testing under severe input conditions, ensuring model resilience and robustness against extreme operational and market challenges.
- Performed exploratory variable research to identify and integrate novel predictors, enhancing the current model suite's performance.

Data Scientist Sep 2021 – Jul 2023 | New York, USA
Smart Energy
- Developed end-to-end Machine Learning models generating actionable business insights, enhancing marketing campaigns for 10,000+ Warm Leads.
- Hands-on data analyses, A/B experiments, and modeling over enormous datasets towards developing insights, recommendations, and presenting results to various stakeholders.

Data Scientist - Research and Analytics Office Apr 2020 – Sep 2021 | Santo Domingo, DR
General Directorate of Internal Taxes
- Created, validated, tested, and tuned Machine Learning algorithms to detect tax noncompliance, manage risk behavior profiles, reduce tax gaps, and build advanced analytics engines.
- Produced analytics to provide stakeholders with data-driven insights for improved tax auditing (the action through which Tax Administrations seek to prevent taxpayers from incurring tax evasion or fraud) by aligning with defined business objectives.
- Analyzed data and provided insights and actionable data-driven recommendations to stakeholders.

Data Analyst - Registry Office Sep 2018 – Apr 2020 | Santo Domingo, DR
General Directorate of Internal Taxes
- Implemented Digital Forms of the National Taxpayer Registry.
- Designed and developed the new engine of automated tax assignation model for the Taxpayers' Registry based on the economic activities and the behavior of the taxpayers.
- Gained knowledge and hands-on experience in Python and usage of machine learning algorithms.

Project Analyst Aug 2017 – Sep 2018 | Santo Domingo, DR
INABIMA
- Establishment of a Risk-based Project Management System.
- Social Impact evaluated by Statistic indicators implementation.

SELECTED PUBLICATIONS

Fernando Martinez, Tao Li, Yingdong Lu, Juntao Chen, "*Stackelberg Coupling of Online Representation Learning and Reinforcement Learning*," In The Fourteenth International Conference on Learning Representations (ICLR 2026).

Fernando Martinez, Tao Li, Yingdong Lu, Juntao Chen, "*In-Context Reinforcement Learning via Communicative World Models*," In The 35th International Joint Conferences on Artificial Intelligence (IJCAI 2026).

Fernando Martinez-Lopez, Juntao Chen, Yingdong Lu, "*SPRIG: Stackelberg Perception-Reinforcement Learning with Internal Game Dynamics*," Presented in AAAI 2025 Workshop on Planning and Reinforcement Learning (PRL) - Bridging the Gap Between AI Planning and Reinforcement Learning.

Pei-Yuan Zhou, Amane Takeuchi, **Fernando Martinez-Lopez**, Malikeh Ehghaghi, Andrew K. C. Wong, En-Shiun Annie Lee, "*Benchmarking Interpretability in Healthcare Using Pattern Discovery and Disentanglement*," Published in MDPI - Bioengineering, 2025.

Yijun Zhao, Alexander Borelli, **Fernando Martinez**, Haoran Xue, Gary M. Weiss, "*Admissions in the age of AI: detecting AI-generated application materials in higher education*," Nature Publishing Group UK, Scientific Reports, 2024.

Fernando Martinez, Mariyam Mapkar, Ali Alfatemi, Mohamed Rahouti, Yufeng Xin, Kaiqi Xiong, Nasir Ghani, "*Redefining Cyber Resilience with Dual-Space Prototypical Networks for DDoS Attack Detection*," In 33rd International Conference on Computer Communications and Networks (ICCCN 2024).

Elizabeth S. Thrall, **Fernando Martinez Lopez**, Thomas J. Egg, Seung Eun Lee, Joshua Schrier, Yijun Zhao, "*Rediscovering the Particle-in-a-Box: Machine Learning Regression Analysis for Hypothesis Generation in Physical Chemistry*," Journal of Chemical Education, ACS Publications, 2023.

SKILLS

Programming Languages — *Python, R, SQL, Bash, C++, MongoDB,* **Python** — *Core, Debugging, Sklearn, NumPy, Pandas, Matplotlib, Seaborn, BeautifulSoup,* **Data Science, Big Data & Machine Learning** — *PyTorch, TensorFlow, Jax, PySpark, Snowflake, Deep Learning, LLMs, Computer Vision,* **Cloud Computing** — *Google Cloud Platform (GCP), Azure, Microsoft Fabric, Docker, Kubernetes,* **BI** — *Power BI, Tableau, SSIS, ETL,* **Microsoft Office** — *Excel, VBA, PPT, Word, Outlook*

HONORS & AWARDS

AI Summer Research Fellowship 2026
GSAS, Fordham University

AI Summer Research Fellowship 2025
GSAS Fordham University

GSAS Graduate Research Assistanship Award 2022
GSAS, Fordham University

Honors 2021
Quantic School of Business and Technology

Full International Scholarship (Fordham University) 2021
Ministry of Higher Education, Science, and Technology of the Dominican Republic

Full Scholarship 2021
Quantic School of Business and Technology

Summa Cum Laude 2018
Santo Domingo Institute of Technology

Meritorious Scholarship 2014
Santo Domingo Institute of Technology
Awarded for top 1% program

ACADEMIC SERVICE & VOLUNTEERING

Conference Reviewer
2026: *CDC, NeurIPS, ICML, AAAI*; 2027: *ACC*

INFOCOM 2025 - NGResNet
Reviewer

TECHO Santo Domingo, Dominican Republic
Volunteer